

Is everything a Niche model?

Tanya Strydom ¹; Xiaoxiao Li ¹; Laura Landon Blake ¹; Andrew P. Beckerman ¹; Dylan Z. Childs ¹

Abstract: TODO

Keywords: food webs

1 Introduction

The Niche Model Hegemony: Establish how the Niche Model (and its topological derivatives like the Cascade model) has historically served as the structural prior or template for most network theory.

The Philosophy Gap: Contrast structural/topological models (which prioritise species-agnostic link distributions) with realised/dynamic models like ATN and ADBM (which prioritise biological mechanisms like bioenergetics and optimal foraging). Additionally also including the MaxEnt model which is a structural model but is based on statistical mechanics rather than the niche template.

The Research Question: Do these different philosophical starting points (mechanical vs. statistical) lead to the same conclusions about community stability, or does the choice of template fundamentally bias our understanding of network dynamics?.

2 Methods

2.1 Network generation

We generated food webs using six established generative models, spanning structural (Niche, Cascade, Random, MaxEnt) and trait-based approaches (Latent Trait Model, Allometric Trophic Network, Allometric Diet Breadth Model). All simulations were implemented in Julia using FoodWebTools and custom extensions.

To isolate model effects from input variation, all trait-based models were parameterised using an identical species pool at each replicate. Species richness was set to range between **XXX**, and for each iteration we sampled (i) a target basal fraction ($U(0.1, 0.3)$), (ii) metabolic classes consistent with this fraction, and (iii) body masses from class-specific log-uniform distributions. These inputs were shared across the Latent Trait, ATN, and ADBM models. Structural models were instead parameterised by a target connectance drawn from $U(0.05, 0.3)$ where relevant.

Networks were generated using a rejection-sampling procedure to enforce comparable emergent properties. A network was retained only if its realised basal fraction and connectance fell within the target ranges (0.1–0.3 and 0.05–0.3, respectively). For trait-based models, networks were accepted only when all three models successfully generated valid networks from the same species pool. We obtained **100** valid replicates per model, with an upper bound on sampling attempts to prevent non-termination. MaxEnt networks were constructed by sampling joint in- and out-degree distributions consistent with a target number of links, followed by entropy-maximising rewiring under degree constraints. For each network, we recorded the adjacency matrix, model identity, input parameters, emergent structural properties, and (where applicable) species traits.

Table 1: Summary of food web models used in this study. Structural models (Niche, Cascade, MaxEnt) generate network topology based on abstract or statistical rules, while realised/dynamic models (ADBM, ATN, Latent trait) incorporate species-specific traits and biological mechanisms to predict interactions. For each model, we list the main inputs and the key assumptions that govern link formation.

Model	Type	Main Inputs	Key Assumptions
Niche model (Williams & Martinez, 2000)	Structural	Species niche values	Trophic interactions structured by a 1D feeding niche; species consume all prey within a contiguous range; allows cannibalism.
Cascade model (Cohen et al., 1990)	Structural	Species rank	Consumers feed only on species with lower ranks; strictly hierarchical; no loops or omnivory.
MaxEnt model (Banville et al., 2023)	Structural	Species richness, connectance	Links assigned to maximize entropy; captures global network properties without species-specific mechanisms.
Allometric Diet Breadth Model (ADBM) (Petchey et al., 2008)	Realised / Dynamic	Body mass, prey energy content	Consumers maximize energy intake; diet breadth determined by profitability and handling time; allometric scaling governs parameters.
Allometric Trophic Network (ATN) (Brose et al., 2006)	Realised / Dynamic	Body mass	Interactions constrained by body-mass ratios; mechanical size limits structure networks; probability of interaction follows Ricker function.
Latent trait model (Rohr et al., 2010)	Realised / Dynamic	Species traits (<i>e.g.</i> , body mass)	Interactions inferred statistically based on hidden trait correlations; can incorporate probabilistic constraints on links.

2.2 Network structure

All networks were converted to directed interaction networks and analysed using a standardised pipeline. We quantified structural properties across multiple scales, including connectance, trophic level, generality, vulnerability, clustering, centrality, trophic coherence, and path length. These metrics were used to characterise the structural signature of each model.

This could include a table that describes what the metrics ‘mean’ but maybe we can just lean on the ‘usual template’, maybe some of the feature selection stuff.

2.3 Dynamic simulations

To evaluate the dynamical consequences of different generative models, all networks were subjected to a common bioenergetic simulation framework using `EcologicalNetworksDynamics.jl`.

For trait-based models (ATN, ADBM, LTN), species body masses were taken from the generative networks and rescaled relative to the producer with the lowest body mass. These models were then parameterised with their corresponding body masses and standard allometric scaling. For structural models (Niche, Cascade, MaxEnt, Random), where explicit trait information was not available, we assigned body masses using trophic scaling with a predator–prey body-mass ratio of 100, a value commonly used for aquatic food webs. Initial species biomasses were randomly assigned. Dynamics were simulated using a Type III functional response, with a Hill exponent of 2.

Each network was simulated until a steady state was reached, with a minimum simulation time of 2000 time units and a maximum of 5000 time units. Equilibrium properties were calculated over the final 500 time units of each simulation. Species were considered extinct and removed when their biomass fell below 10^{-6} . At equilibrium, both extinct species and disconnected species were removed, yielding the realised post-simulation network.

2.4 Post-simulation analyses

Post-simulation analyses quantified two complementary outcomes: community dynamics and realised network topology at equilibrium.

For community dynamics, we focused on persistence (proportion of species surviving), total community biomass, maximum trophic level, diversity (Shannon, evenness), and local stability metrics (resilience and reactivity). Local stability metrics were calculated using `EcoNetPostProcessing.jl`.

To assess how dynamics reshape network topology, we recalculated all structural metrics for the realised (post-extinction) network topology, enabling direct comparison between initial and emergent structure across models.

2.5 Experimental design

The analysis was designed to isolate the effect of network-generating assumptions by holding constant (i) species richness, (ii) trait distributions (within trait-based models), and (iii) dynamical rules. Differences in structure and dynamics can therefore be attributed directly to the generative model used to reconstruct trophic interactions.

3 Results

Structural Signatures: Report how much variance is explained by the model choice.

The Niche Model Template: Identify if the structural models occupy a central ‘mean space’ or if the dynamic models diverge into non-overlapping regions.

[Figure 1 about here.]

Which models are significantly further from Niche than Niche itself: What is the distance of other networks from the niche and does that distance decrease when at the equilibrium state?

[Figure 2 about here.]

Divergent Stories in Dynamics: Show how/if models with similar global structures (like connectance) can lead to wildly different pathways of change/collapse/straight lines.

[Figure 3 about here.]

4 Discussion

Reconstruction is Not Neutral (or maybe it is): Discuss how models act as structural priors that condition ecological inference.

Template Dependence: Address the core question: Is everything a derivative of the niche model? If dynamic models show unique cascade patterns not captured by the niche model, argue that the niche template may be insufficient for dynamic questions.

Scale-Dependent Robustness: Reflect on why species-level patterns might look similar across models while the fine-grained story of dynamic change is highly model-contingent.

Implications for Theory: Suggest that researchers must align their reconstruction framework with their inferential goals rather than defaulting to a one-size-fits-all niche template.

References

- Banville, F., Gravel, D., & Poisot, T. (2023). What constrains food webs? A maximum entropy framework for predicting their structure with minimal biases. *PLOS Computational Biology*, 19(9), e1011458. <https://doi.org/10.1371/journal.pcbi.1011458>
- Brose, U., Jonsson, T., Berlow, E. L., Warren, P., Banasek-Richter, C., Bersier, L.-F., Blanchard, J. L., Brey, T., Carpenter, S. R., Blandenier, M.-F. C., Cushing, L., Dawah, H. A., Dell, T., Edwards, F., Harper-Smith, S., Jacob, U., Ledger, M. E., Martinez, N. D., Memmott, J., ... Cohen, J. E. (2006). Consumer–resource body-size relationships in natural food webs. *Ecology*, 87(10), 2411–2417. [https://doi.org/10.1890/0012-9658\(2006\)87%5B2411:CBRINF%5D2.0.CO;2](https://doi.org/10.1890/0012-9658(2006)87%5B2411:CBRINF%5D2.0.CO;2)
- Cohen, J. E., Briand, F., & Newman, C. (1990). *Community food webs: Data and theory*. Springer-Verlag. <https://www.springer.com/gp/book/9783642837869>
- Petchey, O. L., Beckerman, A. P., Riede, J. O., & Warren, P. H. (2008). Size, foraging, and food web structure. *Proceedings of the National Academy of Sciences*, 105(11), 4191–4196. <https://doi.org/10.1073/pnas.0710672105>
- Rohr, R., Scherer, H., Kehrli, P., Mazza, C., & Bersier, L.-F. (2010). Modeling food webs: Exploring unexplained structure using latent traits. *The American Naturalist*, 176(2), 170–177. <https://doi.org/10.1086/653667>
- Williams, R. J., & Martinez, N. D. (2000). Simple rules yield complex food webs. *Nature*, 404(6774), 180–183. <https://doi.org/10.1038/35004572>

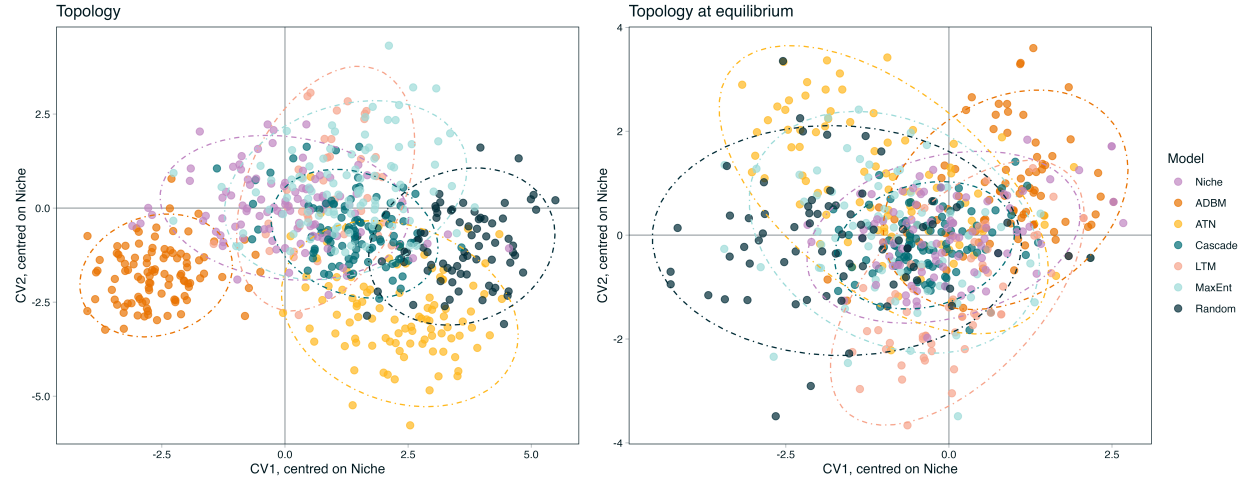


Figure 1: Canonical discriminant analysis (CDA) of ecological network metrics for the seven model types. Each point represents a replicate, and ellipses indicate 95% confidence regions for each model. The second column represents the correlation of the various network metrics with the respective CDA axes. Left is before putting into END and right is post simulation structure (equilibrium topology)

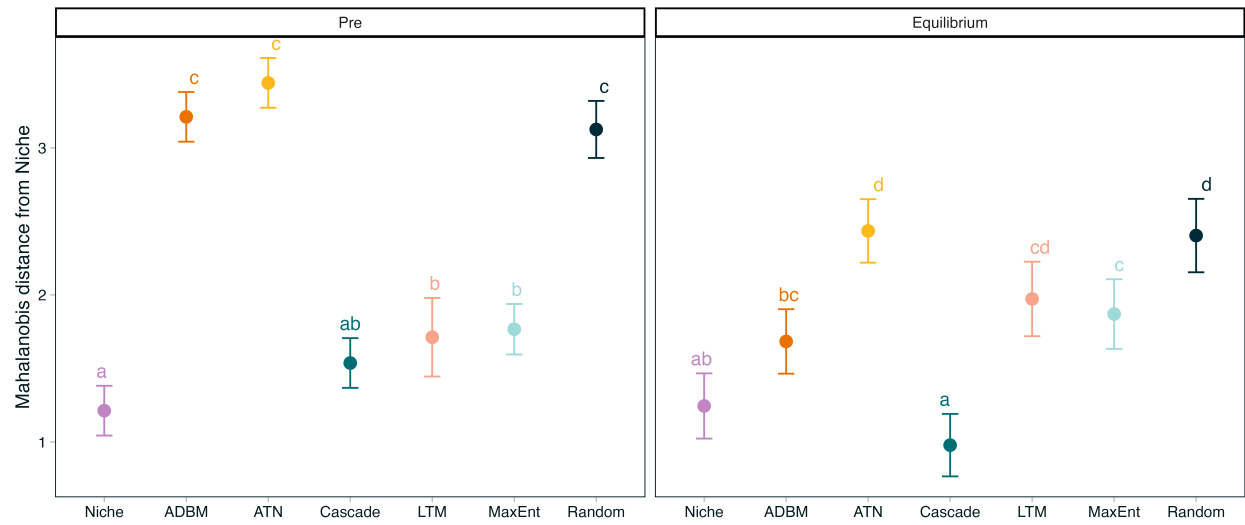


Figure 2: Something something Mahalanobis distance something something

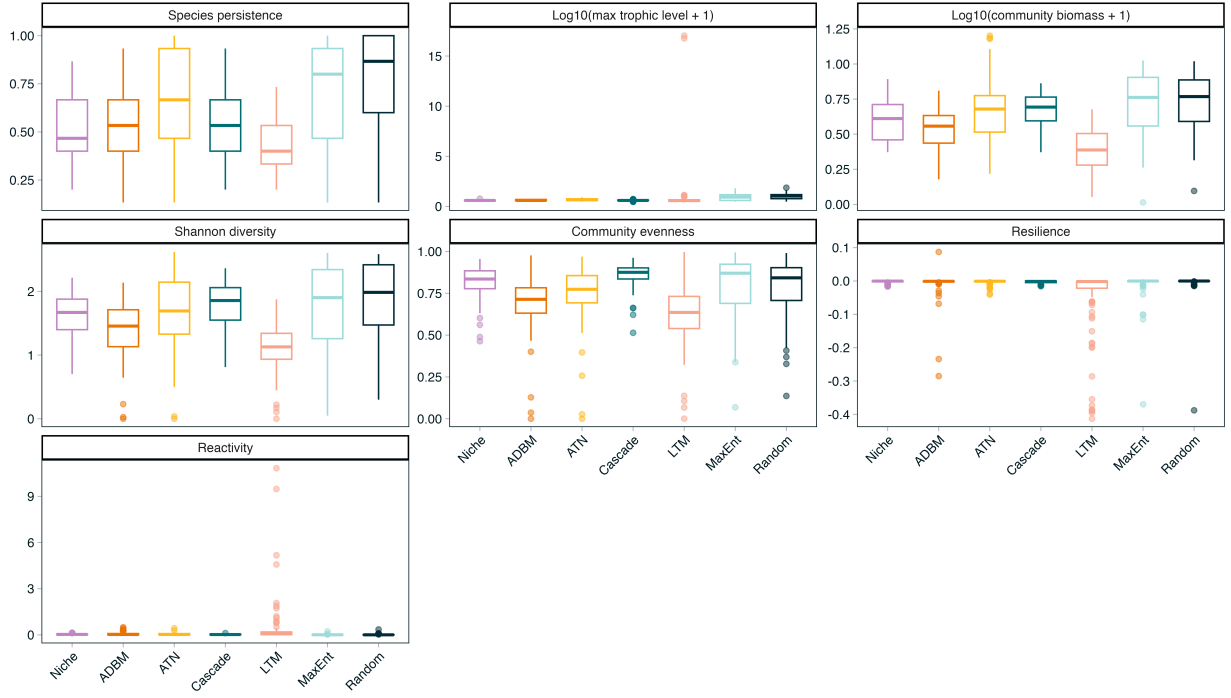


Figure 3: Boxplots showing stability metrics/information about dynamics of the system.